

Postoperative Echography for Optimization of Radiation Dosimetry in Patients with Uveal Melanoma Treated with Plaque Brachytherapy

Elaine M. Binkley, MD,¹ Benjamin A. King, MD,¹ Daniel E. Hyer, PhD,² Asad Javed, MBBS,³ Mohammed M. Milhem, MBBS,³ Connie J. Hinz,¹ Sarah L. Mott, MS,³ H. Culver Boldt, MD¹

Purpose: (1) To describe the technique of postoperative echography to confirm the intended treatment dose to the tumor apex in patients with uveal melanoma treated with plaque brachytherapy. (2) To describe the local tumor control rate and visual outcomes with the brachytherapy strategies used at our institution.

Design: Retrospective review.

Subjects: Three hundred seventy-two consecutive patients with uveal melanoma (small, medium, and large) treated with plaque brachytherapy at the University of Iowa from August 2008 to February 2019.

Methods: Patient demographics and tumor characteristics were recorded for each patient. Patients with posterior tumors treated with plaque brachytherapy ($n = 355$) underwent intraoperative ultrasound to confirm plaque placement, and additional postoperative ultrasound on day 1 to 3 postplaque insertion. In cases where intratumor/episcleral plaque edema or hemorrhage shifted the dose to the prescription point to < 85 Gray (Gy), the duration of plaque brachytherapy was increased to compensate. Statistical analysis was performed to compare variables associated with the need for plaque adjustment.

Main Outcomes Measures: Variables associated with plaque dose needing to be recalculated, local tumor control, and visual acuity outcomes.

Results: In 31 (8.3%) cases, postoperative echography showed that the tumor apex had shifted outside the 85 Gy isodose curve, requiring adjustment of the duration of brachytherapy (28 cases) or repositioning of the plaque (3 cases). Collaborative Ocular Melanoma Study tumor size was significantly associated with need to adjust the plaque prescription dose ($P = 0.03$), with large tumors having the highest rate of adjustment. Tumor thickness was larger in cases requiring plaque adjustment compared with those that were not adjusted (median 4.9 mm vs. 3.0 mm, $P < 0.01$). Local tumor control was 99% (95% confidence interval, 97%–100%) at 5 years and 99% (95% confidence interval, 97%–100%) at 10 years. The percentage of patients who had experienced a visual acuity decline of ≥ 3 lines of vision or had $< 20/200$ acuity was 14.9% at 1 year after brachytherapy, 35.3% at 3 years, and 51.6% at 5 years.

Conclusions: Postoperative ultrasound performed on postoperative day 1 to 3 after plaque insertion for patients undergoing brachytherapy for uveal melanoma may result in improved local tumor control, particularly in the setting of thicker or larger tumors.

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For patients with uveal melanoma, the objectives of local therapy are local tumor control, globe conservation, preservation of vision, and a reduction in the development of metastatic disease. The largest randomized study of patients with medium-sized choroidal melanoma treated with plaque brachytherapy, the Collaborative Ocular Melanoma Study (COMS), had a local tumor control rate of 90% at 5 years, and rates of metastasis of 14% and 23% at 5 and 10 years, respectively.^{1,2} Furthermore, the study showed that with this local tumor control rate, there was no difference in the rate of metastasis compared with patients undergoing enucleation, which has resulted in brachytherapy

becoming the standard of care for managing patients with medium-sized choroidal melanomas.² The indications for plaque brachytherapy have since been expanded to include treatment of both small-sized and large-sized tumors.^{3–5}

Since the COMS, refinement of brachytherapy techniques with strategies, such as intraoperative echography for plaque localization and detection of plaque tilt, have resulted in local tumor control rates of up to 98%.^{6–8} Variables such as radiation dosimetry, edema, or hematoma resulting in plaque tilt, tumor location, tumor size, and use of adjuncts such as transpupillary thermotherapy (TTT) can influence local tumor control.^{6,9,10} Optimization of these variables is